```
# Prediction Test based on Regression and Tree based method


import pandas as pd

# Load the datasets
df_044 = pd.read_csv("mnt-oficial-IJ-044.csv")
df_046 = pd.read_csv("mnt-oficial-IJ-046.csv")
df_117 = pd.read_csv("mnt-oficial-IJ-117.csv")
df_118 = pd.read_csv("mnt-oficial-IJ-118.csv")
df_119 = pd.read_csv("mnt-oficial-IJ-119.csv")
df_120 = pd.read_csv("mnt-oficial-IJ-120.csv")
df_121 = pd.read_csv("mnt-oficial-IJ-121.csv")
df_122 = pd.read_csv("mnt-oficial-IJ-122.csv")

# Concatenate the datasets vertically
merged_df = pd.concat(
    [df_044, df_046, df_117, df_118, df_119, df_120, df_121, df_122], ignore_index=True
)

# Display the first 5 rows
print(merged_df.head().to_markdown(index=False, numalign="left", stralign="left"))

# Print the column names and their data types
print(merged_df.info())
```

```
| Cód. Ordem   | Cód. Recurso   | Cód. Produto   | Qtd. Produzida   | Qtd. Refugada   | Qtd. Retrabalhada   | Fator Un.
|:-------------|:---------------|:---------------|:-----------------|:----------------|:--------------------|:----------
| 2680830      | IJ-044         | SA08395        | 868              | 21              | 0                   | 3
| 2680830      | IJ-044         | SA08395        | 1014             | 15              | 0                   | 4
| 2680534      | IJ-044         | SA07321        | 802              | 46              | 0                   | 3
| 2680534      | IJ-044         | SA07321        | 907              | 18              | 0                   | 3
| 2680830      | IJ-044         | SA08395        | 1033             | 16              | 0                   | 5
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2432 entries, 0 to 2431
Data columns (total 16 columns):
 #   Column                                                  Non-Null Count  Dtype
---  ------                                                  --------------  -----
 0   Cód. Ordem                                              2432 non-null   int64
 1   Cód. Recurso                                            2432 non-null   object
 2   Cód. Produto                                            2432 non-null   object
 3   Qtd. Produzida                                          2432 non-null   int64
 4   Qtd. Refugada                                           2432 non-null   int64
 5   Qtd. Retrabalhada                                       2432 non-null   int64
 6   Fator Un.                                               2432 non-null   int64
 7   Cód. Un.                                                2432 non-null   object
 8   Descrição da massa (Composto)                           2432 non-null   object
 9   Consumo de massa no item em (Kg/100pçs)                 2432 non-null   float64
 10  Qtd. Produzida Acumulada Total                          2432 non-null   int64
 11  Qtd. Refugada Acumulada Total                           2432 non-null   int64
 12  Qtd. Retrabalhada Acumulada Total                       2432 non-null   int64
 13  Fator Un. Acumulado Total                               2432 non-null   int64
 14  Consumo de massa no item em (Kg/100pçs) Acumulado Total 2432 non-null   float64
 15  Manutencao                                              2432 non-null   int64
dtypes: float64(2), int64(10), object(4)
memory usage: 304.1+ KB
None
```

```
# Get all unique values from `Cód. Recurso`
unique_values = merged_df["Cód. Recurso"].unique()

# Check the number of unique values in `Cód. Recurso`
if len(unique_values) > 50:
    # If there are too many unique values, sample the top 50
    top_occuring_values = (
        merged_df["Cód. Recurso"].value_counts().head(50).index.tolist()
    )
    print(top_occuring_values)
else:
    # Otherwise print all unique valus in `Cód. Recurso`
    print(unique_values)
```

```
['IJ-044' 'IJ-046' 'IJ-117' 'IJ-118' 'IJ-119' 'IJ-120' 'IJ-121' 'IJ-122']
```

```
# Get all unique values from `Cód. Produto`
unique_values = merged_df["Cód. Produto"].unique()

# Check the number of unique values in `Cód. Produto`
if len(unique_values) > 50:
    # If there are too many unique values, sample the top 50
    top_occuring_values = (
        merged_df["Cód. Produto"].value_counts().head(50).index.tolist()
```

```
    )
    print(top_occuring_values)
else:
    # Otherwise print all unique valus in `Cód. Produto`
    print(unique_values)

# Get all unique values from `Descrição da massa (Composto)`
unique_values = merged_df["Descrição da massa (Composto)"].unique()

# Check the number of unique values in `Descrição da massa (Composto)`
if len(unique_values) > 50:
    # If there are too many unique values, sample the top 50
    top_occuring_values = (
        merged_df["Descrição da massa (Composto)"]
        .value_counts()
        .head(50)
        .index.tolist()
    )
    print(top_occuring_values)
else:
    # Otherwise print all unique valus in `Descrição da massa (Composto)`
    print(unique_values)
```

```
['SA04416', 'SA07320', 'SA07284', 'SA02794', 'SA05883', 'SA02930', 'SA01923', 'SA02961', 'SA07373', 'SA02693', 'SA02938'
 ['N-142/67' 'N148/18' 'N146/9' 'N-158/3' 'P212/1' 'F420/306' 'F403/32'
  'F420/96' 'F420/191' 'F424/85' 'P248/40' 'F420/95' 'F420/134' 'F424/40 '
  'F424/40' 'S315/10' 'F420/293' 'S316/2' 'P250/14' 'F444/1' ' F403/32'
  'N144/83' ' F420/191' 'F424/51' 'F424/50' 'F420/201' 'N-142/44' 'N108/17']
```

```
# Make a copy of the DataFrame
df_encoded = merged_df.copy()

# Apply one-hot encoding to the non-numeric columns
df_encoded = pd.get_dummies(
    df_encoded,
    columns=["Cód. Produto", "Descrição da massa (Composto)", "Cód. Recurso"],
)

# Display the first 5 rows of the transformed DataFrame
print(df_encoded.head().to_markdown(index=False, numalign="left", stralign="left"))

# Print the column names and their data types of the transformed DataFrame
print(df_encoded.info())
```

```
| Cód. Ordem   | Qtd. Produzida   | Qtd. Refugada   | Qtd. Retrabalhada   | Fator Un.   | Cód. Un.   | Consumo de massa
|:-------------|:-----------------|:----------------|:--------------------|:------------|:-----------|:-----------------
| 2680830      | 868              | 21              | 0                   | 3           | PC         | 8.607
| 2680830      | 1014             | 15              | 0                   | 4           | PC         | 9.157
| 2680534      | 802              | 46              | 0                   | 3           | PC         | 4.35
| 2680534      | 907              | 18              | 0                   | 3           | PC         | 5.7
| 2680830      | 1033             | 16              | 0                   | 5           | PC         | 12.407
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2432 entries, 0 to 2431
Columns: 350 entries, Cód. Ordem to Cód. Recurso_IJ-122
dtypes: bool(337), float64(2), int64(10), object(1)
memory usage: 1.0+ MB
None
```

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

# Drop the `Manutencao` column for the target variable
y = df_encoded["Manutencao"]

# Drop the remaining non-numeric columns and the target variable
X = df_encoded.drop(columns=["Manutencao", "Cód. Un."])

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)


# Create a Linear Regression model
model = LinearRegression()

# Fit the model to the training data
model.fit(X_train, y_train)

# Print the model's score
print("Model score:", model.score(X_test, y_test))
```

```
Model score: 0.9851673494016067
```

```
# Convert `Cód. Produto` and `Descrição da massa (Composto)` to string type
merged_df["Cód. Produto"] = merged_df["Cód. Produto"].astype(str)
merged_df["Descrição da massa (Composto)"] = merged_df[
    "Descrição da massa (Composto)"
].astype(str)

# Print the column names and their data types
print(merged_df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2432 entries, 0 to 2431
Data columns (total 16 columns):
 #   Column                                                  Non-Null Count  Dtype
---  ------                                                  --------------  -----
 0   Cód. Ordem                                              2432 non-null   int64
 1   Cód. Recurso                                            2432 non-null   object
 2   Cód. Produto                                            2432 non-null   object
 3   Qtd. Produzida                                          2432 non-null   int64
 4   Qtd. Refugada                                           2432 non-null   int64
 5   Qtd. Retrabalhada                                       2432 non-null   int64
 6   Fator Un.                                               2432 non-null   int64
 7   Cód. Un.                                                2432 non-null   object
 8   Descrição da massa (Composto)                           2432 non-null   object
 9   Consumo de massa no item em (Kg/100pçs)                 2432 non-null   float64
 10  Qtd. Produzida Acumulada Total                          2432 non-null   int64
 11  Qtd. Refugada Acumulada Total                           2432 non-null   int64
 12  Qtd. Retrabalhada Acumulada Total                       2432 non-null   int64
 13  Fator Un. Acumulado Total                               2432 non-null   int64
 14  Consumo de massa no item em (Kg/100pçs) Acumulado Total 2432 non-null   float64
 15  Manutencao                                              2432 non-null   int64
dtypes: float64(2), int64(10), object(4)
memory usage: 304.1+ KB
None
```

```
# Apply one-hot encoding to the non-numeric columns
df = pd.get_dummies(
    merged_df,
    columns=[
        "Cód. Produto",
        "Descrição da massa (Composto)",
        "Cód. Recurso",
    ],
)

# Display the first 5 rows of the transformed DataFrame
print(df.head().to_markdown(index=False, numalign="left", stralign="left"))

# Print the column names and their data types of the transformed DataFrame
print(df.info())
```

```
| Cód. Ordem   | Qtd. Produzida   | Qtd. Refugada   | Qtd. Retrabalhada   | Fator Un.   | Cód. Un.   | Consumo de massa
|:-------------|:-----------------|:----------------|:--------------------|:------------|:-----------|:-----------------
| 2680830      | 868              | 21              | 0                   | 3           | PC         | 8.607
| 2680830      | 1014             | 15              | 0                   | 4           | PC         | 9.157
| 2680534      | 802              | 46              | 0                   | 3           | PC         | 4.35
| 2680534      | 907              | 18              | 0                   | 3           | PC         | 5.7
| 2680830      | 1033             | 16              | 0                   | 5           | PC         | 12.407
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2432 entries, 0 to 2431
Columns: 350 entries, Cód. Ordem to Cód. Recurso_IJ-122
dtypes: bool(337), float64(2), int64(10), object(1)
memory usage: 1.0+ MB
None
```

```
# Drop the `Manutencao` column for the target variable
y = df["Manutencao"]

# Drop the remaining non-numeric columns and the target variable
X = df.drop(columns=["Manutencao", "Cód. Un."])

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
import numpy as np

X_train = X_train.to_numpy()
y_train = y_train.to_numpy()
```

```
X_test = X_test.to_numpy()
y_test = y_test.to_numpy()


# Create a Linear Regression model
model = LinearRegression()

# Fit the model to the training data
model.fit(X_train, y_train)

# Print the model's score
print("Model score:", model.score(X_test, y_test))
```

```
Model score: 0.9851673494016067
```

```
print("Model score:", model.score(X_train, y_train))
```

```
Model score: 0.9935526970887113
```

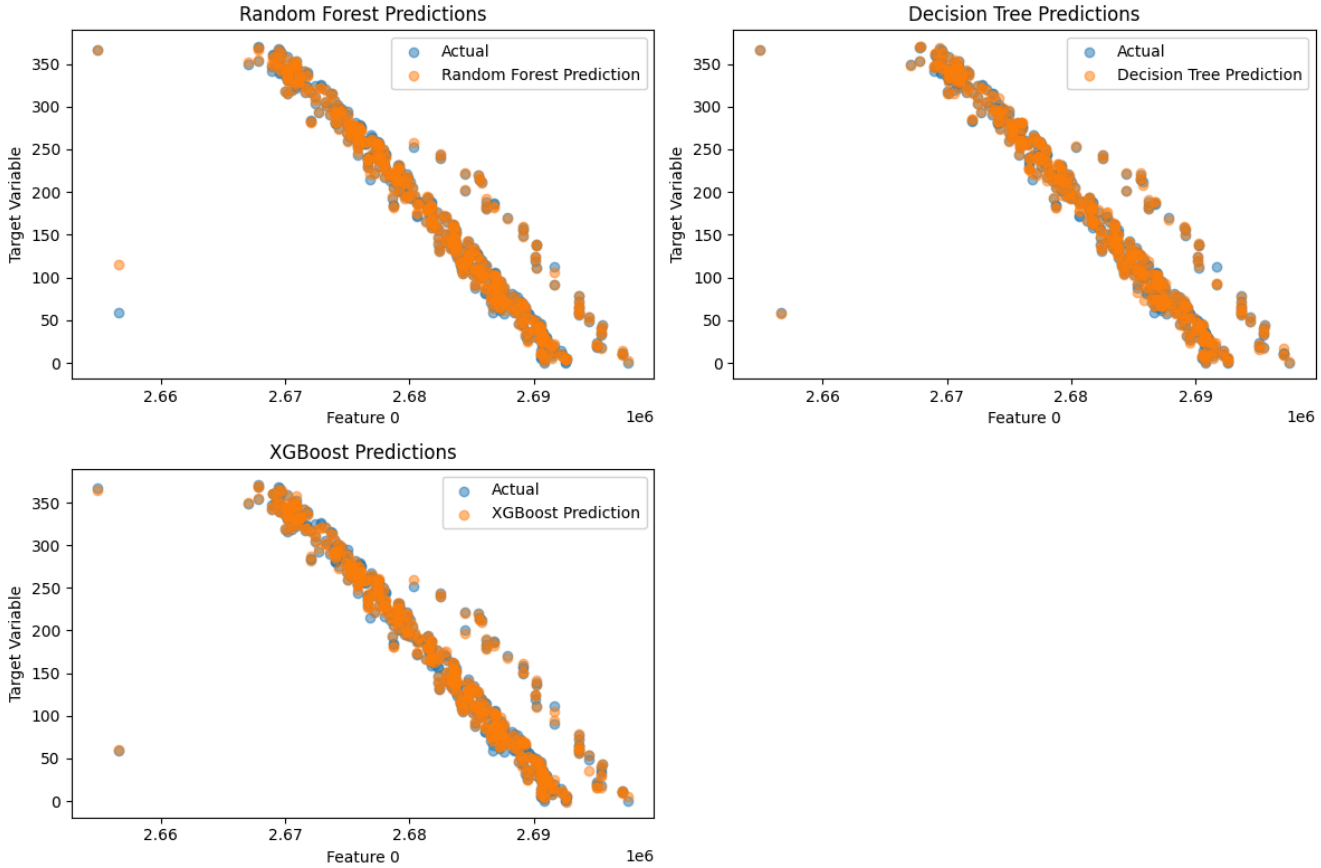
```
for i in range(10):
    random_number = np.random.randint(0, len(X_train))
    prediction = model.predict(X_train[random_number].reshape(1, -1))
    print(f"""Modelo: {prediction}, Real: {y_train[random_number].reshape(-1)}""")
```

```
Modelo: [238.64987826], Real: [228]
Modelo: [83.91438094], Real: [77]
Modelo: [78.17576201], Real: [77]
Modelo: [107.15436422], Real: [107]
Modelo: [239.20286223], Real: [237]
Modelo: [166.20295911], Real: [164]
Modelo: [198.52308653], Real: [202]
Modelo: [167.02859777], Real: [165]
Modelo: [100.91332138], Real: [108]
Modelo: [118.4481572], Real: [118]
```

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.tree import DecisionTreeRegressor
import xgboost as xgb
from sklearn.metrics import mean_squared_error, r2_score
import matplotlib.pyplot as plt


models = {
    "Random Forest": RandomForestRegressor(random_state=42),
    "Decision Tree": DecisionTreeRegressor(random_state=42),
    "XGBoost": xgb.XGBRegressor(
        random_state=42, objective="reg:squarederror"
    ),  # Important for regression
}

results = {}

for name, model in models.items():
    print(f"Training and evaluating {name}...")
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    # Evaluate the model
    mse = mean_squared_error(y_test, y_pred)
    r2 = r2_score(y_test, y_pred)

    results[name] = {"model": model, "mse": mse, "r2": r2, "y_pred": y_pred}

# Compare the models
print("\nModel Comparison:")
for name, metrics in results.items():
    print(f"{name}:")
    print(f"  MSE: {metrics['mse']:.4f}")
    print(f"  R-squared: {metrics['r2']:.4f}")

# Plotting:
plt.figure(figsize=(12, 8))
for i, (name, metrics) in enumerate(results.items()):
    plt.subplot(2, 2, i + 1)
    plt.scatter(X_test[:, 0], y_test, label="Actual", alpha=0.5)  # Use array indexing
    plt.scatter(
        X_test[:, 0], metrics["y_pred"], label=name + " Prediction", alpha=0.5
    )  # Use array indexing
    plt.xlabel("Feature 0")  # Or a more descriptive name if you know it
```

```
    plt.ylabel("Target Variable")
    plt.title(name + " Predictions")
    plt.legend()
plt.tight_layout()
plt.show()
```

```
Training and evaluating Random Forest...
Training and evaluating Decision Tree...
Training and evaluating XGBoost...

Model Comparison:
Random Forest:
  MSE: 11.5044
  R-squared: 0.9990
Decision Tree:
  MSE: 12.4271
  R-squared: 0.9989
XGBoost:
  MSE: 9.0767
  R-squared: 0.9992
```




```
# Example: Accessing the best model (based on R-squared)
best_model_name = max(results, key=lambda k: results[k]["r2"])
best_model = results[best_model_name]["model"]

print(f"\nBest Model: {best_model_name}")
```

```
Best Model: XGBoost
```

```
# Find the best model based on a chosen metric (e.g., R-squared)
best_model_name = max(results, key=lambda k: results[k]["r2"])
best_model_metrics = results[best_model_name]
# Print the best model's metrics
print(f"\nBest Model: {best_model_name}")
print("Metrics for the best model:")
for metric_name, metric_value in best_model_metrics.items():
    if metric_name != "model":  # Exclude the model object itself
        print(f"  {metric_name}: {metric_value}")
```

```
Best Model: XGBoost
Metrics for the best model:
  mse: 9.076658264742065
  r2: 0.9991755485534668
  y_pred: [113.782074     3.7019563  258.55304     75.792725    19.291529
 181.5844      21.389658   108.13632    330.14987    133.69867
 207.7275     289.0642      16.059711    13.977041   109.77427
   5.877147   162.50252    139.13388    264.83533    292.48282
 196.67365    293.44702     75.131714   183.31749     11.784785
 273.7389     259.16037    319.1772     104.94397    193.8949
  34.366234   116.50461     86.71154     75.85347    225.5408
 121.68816    148.52072    125.059586   348.4633      43.444664
 161.42973    214.69304    266.41397    196.8348     235.37396
 339.82504    253.94519    350.90936     78.97202    303.18607
 131.674      215.32953    320.3026     221.68901      7.249278
 139.07362     37.34137     74.89355    138.65068    190.8504
  77.60436    127.05019    360.4927     112.24115    105.23375
  43.272602    42.776947   256.24826    235.9294     154.00363
 173.4966      65.455284   313.59485    281.72717    232.29132
  63.311787   179.09738     64.49506    152.48628     22.739319
 241.52983    259.2507     127.72418    319.31348    247.57745
  85.14316    206.78275     54.311665   142.99376    115.59186
 336.88086    268.16287    243.0465     355.17044    129.40828
 165.94554    259.97922    228.81335    277.0686      89.404366
  75.73162    219.54477    129.36389     44.439903   135.81984
   4.4074507   64.542786    55.126965   137.15425     16.258644
  35.21901    246.13582    258.1617     318.74625    337.1337
 258.60367     24.573778   253.99309    138.45384     47.064766
 266.73224     60.513416   122.94764    166.30077    198.72144
 109.77779     76.363594   342.06387    357.1405      84.09083
  15.780273     3.499348   263.02924    319.639      105.883705
 350.96866    162.00711    342.4253     366.3377     335.60867
 358.42538    151.48166     70.76533    168.74612    329.934
 273.79248    150.65367     98.63904    139.91098    360.87662
 228.2339      12.909806    76.88112    274.60815     79.01036
  49.90143     75.24854     70.44757    231.60251    343.21225
 154.1567     221.10329    319.28717    115.4739      47.514908
 150.5906     236.36375    323.67914    217.65329    331.51395
 112.215546   242.79504    337.71494    103.09578    295.8191
 305.1444       5.25188    163.16861    114.59289    349.1325
 212.4918      14.676428   128.00404    245.99693     72.80279
 112.136154    26.52442     32.154312   187.01053    273.60825
  67.53037    258.18475    102.261375     4.021893   346.32834
 301.37167     65.26604    165.66447    180.30504    275.70322
  29.01889    260.29678     75.76849     60.64133    170.06635
 212.84941    217.3428     239.35918    222.49422     34.202595
  19.648073   323.33972    277.61124    332.80066    333.95407
  -0.57927567 254.9324     343.97345    219.83139     10.7052145
 220.55675    296.441       87.2615      30.912518    86.011604
  46.999863   347.5677      29.451324   346.95853     92.968155
 258.8297     175.28717     34.748432   203.13757     24.771523
  24.873005   187.55583      6.6298366  186.33864     51.47057
   7.5035415   72.19958    320.53036    257.69766     71.16225
 151.44894     10.416637   249.68314     52.816463   342.4543
 336.4828     198.02219     99.80063    133.43077    177.52786
  97.35642     34.45946    176.2244     105.052574    32.943275
 333.18994    195.35231     88.390305   299.86343    105.1732
```

```
for name, model_data in results.items():
    model = model_data["model"]
    predictions = model.predict(X_train)
    print(f"\nPredictions using {name} on random data:")
    print(predictions)
```

```
Predictions using Random Forest on random data:
[ 76.5   73.9  324.49 ...  31.53 206.62 351.95]

Predictions using Decision Tree on random data:
[ 77.  73. 327. ...  32. 206. 352.]

Predictions using XGBoost on random data:
[ 76.46936   75.90217  324.68292  ...  31.163788 206.32513  350.378   ]
```

```
print(y_train)
```

```
[ 77  73 327 ...  32 206 352]
```

```
for name, model_data in results.items():
    model = model_data["model"]
    predictions = model.predict(X_test[:50])
    print(f"\nPredictions using {name} on random data:")
    print(predictions)
```

```
Predictions using Random Forest on random data:
[110.66   3.85 260.69  71.5   20.56 182.48  23.23 109.58 328.38 130.68
 204.76 291.51  19.24  13.92 111.7    6.55 161.54 137.64 261.51 291.54
 192.5  293.04  71.97 183.02  14.21 270.84 257.3  317.74 104.08 194.37
  34.22 114.98  83.76  72.26 226.6  119.48 147.06 130.05 353.7   43.27
 159.29 215.37 267.04 202.69 239.37 343.58 255.18 351.96  79.53 302.97]

Predictions using Decision Tree on random data:
[112.   3. 261.  72.  23. 180.  27. 109. 331. 132. 206. 296.  16.  13.
 109.   7. 163. 136. 265. 293. 196. 293.  72. 180.  17. 273. 258. 318.
 104. 194.  28. 113.  85.  75. 227. 116. 144. 129. 355.  43. 160. 214.
 265. 202. 240. 345. 257. 351.  79. 303.]

Predictions using XGBoost on random data:
[113.782074    3.7019563 258.55304    75.792725   19.291529  181.5844
  21.389658  108.13632   330.14987   133.69867   207.7275    289.0642
  16.059711   13.977041  109.77427     5.877147  162.50252   139.13388
 264.83533   292.48282   196.67365   293.44702    75.131714  183.31749
  11.784785  273.7389    259.16037   319.1772    104.94397   193.8949
  34.366234  116.50461    86.71154    75.85347   225.5408    121.68816
 148.52072   125.059586  348.4633     43.444664  161.42973   214.69304
 266.41397   196.8348    235.37396   339.82504   253.94519   350.90936
  78.97202   303.18607  ]
```

```
y_test[:50]
```

```
array([111,   2, 262,  68,  13, 179,  21, 108, 326, 131, 206, 295,  19,
        13, 110,   8, 162, 137, 264, 290, 195, 292,  73, 181,  12, 271,
       259, 317, 105, 193,  30, 114,  83,  74, 228, 121, 148, 131, 354,
        44, 159, 213, 265, 201, 236, 340, 256, 352,  80, 301])
```

```
import pandas as pd
from sklearn.model_selection import (
    train_test_split,
    cross_val_score,
    KFold,
    GridSearchCV,
)
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import (
    RandomForestClassifier,
    GradientBoostingClassifier,
)  # Added GradientBoosting
from sklearn.metrics import roc_curve, auc, confusion_matrix, classification_report
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

```
# 1. Decision Tree
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train)
y_pred_dt = dt_model.predict(X_test)
y_prob_dt = dt_model.predict_proba(X_test)[:, 1]
```

```
# 2. Random Forest (as you had before, but I'll include it for comparison)
rf_model = RandomForestClassifier(random_state=42)
rf_model.fit(X_train, y_train)
y_pred_rf = rf_model.predict(X_test)
y_prob_rf = rf_model.predict_proba(X_test)[:, 1]
```

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from sklearn.ensemble import GradientBoostingRegressor, RandomForestRegressor

# Feature importance
tree_models = {
    "Decision Tree Regression": DecisionTreeRegressor(random_state=42),
    "Random Forest Regression": RandomForestRegressor(random_state=42),
    "Gradient Boosting Regression": GradientBoostingRegressor(random_state=42),
}
for model_name, model in tree_models.items():
    model.fit(X_train, y_train)
    importance = model.feature_importances_
    print(f"{model_name} - Feature Importance:")
    for i, feature_name in enumerate(X.columns):
        print(f"{feature_name}: {importance[i]:.4f}")
```

```
Decision Tree Regression - Feature Importance:
Cód. Ordem: 0.8578
```

```
Qtd. Produzida: 0.0001
Qtd. Refugada: 0.0001
Qtd. Retrabalhada: 0.0000
Fator Un.: 0.0000
Consumo de massa no item em (Kg/100pçs): 0.0001
Qtd. Produzida Acumulada Total: 0.0059
Qtd. Refugada Acumulada Total: 0.0096
Qtd. Retrabalhada Acumulada Total: 0.0000
Fator Un. Acumulado Total: 0.1234
Consumo de massa no item em (Kg/100pçs) Acumulado Total: 0.0010
Cód. Produto_10436: 0.0000
Cód. Produto_11974: 0.0000
Cód. Produto_12139: 0.0000
Cód. Produto_12693: 0.0000
Cód. Produto_12973: 0.0000
Cód. Produto_13444: 0.0000
Cód. Produto_15570: 0.0000
Cód. Produto_15958: 0.0000
Cód. Produto_17395: 0.0000
Cód. Produto_18433: 0.0000
Cód. Produto_18454: 0.0000
Cód. Produto_27298: 0.0000
Cód. Produto_SA00013: 0.0000
Cód. Produto_SA00049: 0.0000
Cód. Produto_SA00063: 0.0000
Cód. Produto_SA00181: 0.0000
Cód. Produto_SA00224: 0.0000
Cód. Produto_SA00228: 0.0000
Cód. Produto_SA00272: 0.0000
Cód. Produto_SA00421: 0.0000
Cód. Produto_SA00436: 0.0000
Cód. Produto_SA00484: 0.0000
Cód. Produto_SA00497: 0.0000
Cód. Produto_SA00498: 0.0000
Cód. Produto_SA00501: 0.0000
Cód. Produto_SA00504: 0.0000
Cód. Produto_SA00618: 0.0000
Cód. Produto_SA00734: 0.0000
Cód. Produto_SA00775: 0.0000
Cód. Produto_SA00793: 0.0000
Cód. Produto_SA00803: 0.0000
Cód. Produto_SA00834: 0.0000
Cód. Produto_SA00880: 0.0000
Cód. Produto_SA00911: 0.0000
Cód. Produto_SA00912: 0.0000
Cód. Produto_SA00913: 0.0000
Cód. Produto_SA00946: 0.0000
Cód. Produto_SA00995: 0.0000
Cód. Produto_SA01366: 0.0000
Cód. Produto_SA01422: 0.0000
Cód. Produto_SA01495: 0.0000
Cód. Produto_SA01535: 0.0000
Cód. Produto_SA01550: 0.0000
Cód. Produto_SA01556: 0.0000
Cód. Produto_SA01660: 0.0000
Cód. Produto_SA01684: 0.0000
```

```
def evaluate_model(model, y_true, y_pred, y_prob, model_name):
    print(f"--- {model_name} Evaluation ---")

    # ROC Curve
    fpr, tpr, _ = roc_curve(y_true, y_prob)
    roc_auc = auc(fpr, tpr)
    plt.figure(figsize=(8, 6))
    plt.plot(
        fpr, tpr, color="darkorange", lw=2, label=f"ROC curve (area = {roc_auc:.2f})"
    )
    plt.plot([0, 1], [0, 1], color="navy", lw=2, linestyle="--")
    plt.xlim([0.0, 1.0])
    plt.ylim([0.0, 1.05])
    plt.xlabel("False Positive Rate")
    plt.ylabel("True Positive Rate")
    plt.title(f"ROC Curve - {model_name}")
    plt.legend(loc="lower right")
    plt.show()

    # Confusion Matrix
    cm = confusion_matrix(y_true, y_pred)
    plt.figure(figsize=(6, 6))
    sns.heatmap(
        cm,
        annot=True,
        fmt="d",
        cmap="Blues",
        cbar=False,
        xticklabels=np.unique(y),
        yticklabels=np.unique(y),
```

```
    )
    plt.xlabel("Predicted Label")
    plt.ylabel("True Label")
    plt.title(f"Confusion Matrix - {model_name}")
    plt.show()

    # Classification Report
    print(classification_report(y_true, y_pred))

    # Cross-validation (example for Random Forest, adapt as needed)
    if model_name == "Random Forest":  # Only do CV for RF in this example
        kfold = KFold(n_splits=5, shuffle=True, random_state=42)
        cv_scores = cross_val_score(
            model, X_train, y_train, cv=kfold, scoring="accuracy"
        )
        print("Cross-validation scores:", cv_scores)
        print("Mean CV accuracy:", cv_scores.mean())


from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    roc_auc_score,
    confusion_matrix,
)


def evaluate_model(model, y_true, y_pred, y_prob, model_name):
    """Evaluates a classification model and prints the results."""

    print(f"--- {model_name} ---")
    print("Accuracy:", accuracy_score(y_true, y_pred))
    print("Precision:", precision_score(y_true, y_pred))
    print("Recall:", recall_score(y_true, y_pred))
    print("F1-score:", f1_score(y_true, y_pred))
    if y_prob is not None:
        print("ROC-AUC:", roc_auc_score(y_true, y_prob))
    print("Confusion Matrix:\n", confusion_matrix(y_true, y_pred))
    print("\n")


# Define and train models
dt_model = DecisionTreeClassifier(random_state=42)
rf_model = RandomForestClassifier(random_state=42)
gb_model = GradientBoostingClassifier(random_state=42)

dt_model.fit(X_train, y_train)
rf_model.fit(X_train, y_train)
gb_model.fit(X_train, y_train)

# Make predictions
y_pred_dt = dt_model.predict(X_test)
y_prob_dt = dt_model.predict_proba(X_test)[:, 1]
y_pred_rf = rf_model.predict(X_test)
y_prob_rf = rf_model.predict_proba(X_test)[:, 1]
y_pred_gb = gb_model.predict(X_test)
y_prob_gb = gb_model.predict_proba(X_test)[:, 1]

# Define evaluate_model function (as shown earlier)

# Evaluate all models
evaluate_model(dt_model, y_test, y_pred_dt, y_prob_dt, "Decision Tree")
```

```
---------------------------------------------------------------------------
KeyboardInterrupt                         Traceback (most recent call last)
Cell In[89], line 8
      6 dt_model.fit(X_train, y_train)
      7 rf_model.fit(X_train, y_train)
----> 8 gb_model.fit(X_train, y_train)
     10 # Make predictions
     11 y_pred_dt = dt_model.predict(X_test)

File ~/opt/anaconda3/envs/oxaala/lib/python3.9/site-packages/sklearn/base.py:1473, in _fit_context.<locals>.decorator.
<locals>.wrapper(estimator, *args, **kwargs)
   1466     estimator._validate_params()
   1468 with config_context(
   1469     skip_parameter_validation=(
   1470         prefer_skip_nested_validation or global_skip_validation
   1471     )
   1472 ):
-> 1473     return fit_method(estimator, *args, **kwargs)

File ~/opt/anaconda3/envs/oxaala/lib/python3.9/site-packages/sklearn/ensemble/_gb.py:783, in
BaseGradientBoosting.fit(self, X, y, sample_weight, monitor)
    780     self._resize_state()
    782 # fit the boosting stages
--> 783 n_stages = self._fit_stages(
    784     X_train,
    785     y_train,
    786     raw_predictions,
    787     sample_weight_train,
    788     self._rng,
    789     X_val,
    790     y_val,
    791     sample_weight_val,
    792     begin_at_stage,
    793     monitor,
    794 )
    796 # change shape of arrays after fit (early-stopping or additional ests)
    797 if n_stages != self.estimators_.shape[0]:

File ~/opt/anaconda3/envs/oxaala/lib/python3.9/site-packages/sklearn/ensemble/_gb.py:879, in
BaseGradientBoosting._fit_stages(self, X, y, raw_predictions, sample_weight, random_state, X_val, y_val,
sample_weight_val, begin_at_stage, monitor)
    872         initial_loss = factor * self._loss(
    873             y_true=y_oob_masked,
    874             raw_prediction=raw_predictions[~sample_mask],
    875             sample_weight=sample_weight_oob_masked,
    876         )
    878 # fit next stage of trees
--> 879 raw_predictions = self._fit_stage(
    880     i,
    881     X,
    882     y,
    883     raw_predictions,
    884     sample_weight,
    885     sample_mask,
    886     random_state,
    887     X_csc=X_csc,
    888     X_csr=X_csr,
    889 )
    891 # track loss
    892 if do_oob:

File ~/opt/anaconda3/envs/oxaala/lib/python3.9/site-packages/sklearn/ensemble/_gb.py:490, in
BaseGradientBoosting._fit_stage(self, i, X, y, raw_predictions, sample_weight, sample_mask, random_state, X_csc,
X_csr)
    487     sample_weight = sample_weight * sample_mask.astype(np.float64)
    489 X = X_csc if X_csc is not None else X
--> 490 tree.fit(
    491     X, neg_g_view[:, k], sample_weight=sample_weight, check_input=False
    492 )
    494 # update tree leaves
    495 X_for_tree_update = X_csr if X_csr is not None else X

File ~/opt/anaconda3/envs/oxaala/lib/python3.9/site-packages/sklearn/base.py:1473, in _fit_context.<locals>.decorator.
<locals>.wrapper(estimator, *args, **kwargs)
   1466     estimator._validate_params()
   1468 with config_context(
   1469     skip_parameter_validation=(
   1470         prefer_skip_nested_validation or global_skip_validation
   1471     )
   1472 ):
-> 1473     return fit_method(estimator, *args, **kwargs)

File ~/opt/anaconda3/envs/oxaala/lib/python3.9/site-packages/sklearn/tree/_classes.py:1377, in
DecisionTreeRegressor.fit(self, X, y, sample_weight, check_input)
   1347 @_fit_context(prefer_skip_nested_validation=True)
   1348 def fit(self, X, y, sample_weight=None, check_input=True):
   1349     """Build a decision tree regressor from the training set (X, y).
   1350
   1351     Parameters
   (...)
```

```
   1374         Fitted estimator.
   1375     """
-> 1377     super()._fit(
   1378         X,
   1379         y,
   1380         sample_weight=sample_weight,
   1381         check_input=check_input,
   1382     )
   1383     return self

File ~/opt/anaconda3/envs/oxaala/lib/python3.9/site-packages/sklearn/tree/_classes.py:472, in
BaseDecisionTree._fit(self, X, y, sample_weight, check_input, missing_values_in_feature_mask)
    461 else:
    462     builder = BestFirstTreeBuilder(
    463         splitter,
    464         min_samples_split,
   (...)
    469         self.min_impurity_decrease,
```

```
evaluate_model(rf_model, y_test, y_pred_rf, y_prob_rf, "Random Forest")


evaluate_model(gb_model, y_test, y_pred_gb, y_prob_gb, "Gradient Boosting")


from sklearn.model_selection import train_test_split

# Separate features and target
X = merged_df[
    [
        "Qtd. Produzida",
        "Qtd. Refugada",
        "Qtd. Retrabalhada",
        "Fator Un.",
        "Consumo de massa no item em (Kg/100pçs)",
    ]
]
y = merged_df["Manutencao"]

# Split data into train and test sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Display the shapes of the train and test sets
print(f"X_train shape: {X_train.shape}")
print(f"X_test shape: {X_test.shape}")
print(f"y_train shape: {y_train.shape}")
print(f"y_test shape: {y_test.shape}")


from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.metrics import mean_squared_error

# Initialize the models
models = {
    "Linear Regression": LinearRegression(),
    "Decision Tree Regression": DecisionTreeRegressor(random_state=42),
    "Random Forest Regression": RandomForestRegressor(random_state=42),
    "Gradient Boosting Regression": GradientBoostingRegressor(random_state=42),
}

# Train and predict with each model
predictions = {}
for model_name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    predictions[model_name] = y_pred

# Evaluate each model using RMSE
rmse_results = {}
for model_name, y_pred in predictions.items():
    rmse = mean_squared_error(y_test, y_pred, squared=False)
    rmse_results[model_name] = rmse

# Display the RMSE results
for model_name, rmse in rmse_results.items():
    print(f"{model_name} RMSE: {rmse}")


import matplotlib.pyplot as plt
from sklearn.model_selection import GridSearchCV
```